

READING PASSAGE 3

The Internet of Things in Smart Cities

A The study of the internet of things in smart cities has undergone a remarkable transformation over the past several decades. What was once a niche area of academic inquiry has developed into a rich and diverse field of research, drawing on insights from a wide range of disciplines and theoretical traditions.

B The origins of contemporary research in this area can be traced to pioneering studies conducted during the mid-twentieth century. These early investigations, though constrained by the technological and methodological limitations of their era, established foundational principles that continue to inform current scholarship. Successive generations of researchers have built upon these foundations, progressively refining theoretical models and expanding the empirical evidence base through increasingly sophisticated research designs.

C At the core of current understanding lies a set of interrelated mechanisms that collectively explain the observed patterns and outcomes. First, structural factors create the underlying conditions within which individual and collective action takes place. These structural dimensions include economic incentives, institutional frameworks, and cultural norms that shape behaviour in ways that are often not immediately apparent to the actors themselves. Second, dynamic feedback processes amplify or dampen initial effects, generating patterns of change that can diverge substantially from what linear models would predict.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G The evidence reviewed above demonstrates both the substantial achievements of research in this field and the significant work that remains to be done. As new tools and theoretical perspectives become available, and as the phenomena themselves continue to evolve, the research community will need to maintain both rigour in its methods and openness to new ideas. The stakes are high, for the questions addressed by this research bear directly on some of the most important challenges facing humanity in the twenty-first century.

Question 1

Choose the correct letter, A, B, C or D.

1 The meta-analyses cited in the passage reveal that:

- A. minimal impact
- B. creating underlying conditions for action
- C. determining individual choices directly
- D. irrelevant to outcomes

Question 2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

2 The author notes that researchers have called for more _____ approaches to investigation.

Question 3

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

3 Early researchers had access to the same technologies as modern scholars.

Questions 4–5

Choose the correct letter, A, B, C or D.

4 The author suggests that early studies in this field were limited by:

- A. too much data collection
- B. the generalisability of findings across contexts
- C. insufficient government funding
- D. overly complex statistical methods

5 What does the passage identify as a significant challenge to current understanding?

- A. inconsistent results across cultures
- B. a consistent pattern across diverse contexts
- C. no measurable effects
- D. effects limited to Western countries

Question 6

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

6 The passage suggests that structural factors are less important than individual choices.

Questions 7–8

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

7 The passage identifies power _____ as a key factor at the organisational level.

8 According to the text, cognitive processes shape the _____ that people make.

Question 9

Choose the correct letter, A, B, C or D.

9 The meta-analyses cited in the passage reveal that:

- A. weak effects across studies
- B. moderate to large effect sizes with consistency
- C. contradictory findings
- D. results only in specific populations

Question 10

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

10 The author believes that all research questions in this field have been answered.

Question 11

Choose the correct letter, A, B, C or D.

11 The passage indicates that critics have raised concerns about:

- A. inconsistent results across cultures
- B. a consistent pattern across diverse contexts
- C. no measurable effects
- D. effects limited to Western countries

Question 12

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

12 The landmark study discussed in the passage involved participants from multiple countries.

Question 13

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

13 Case studies have documented the effects of _____ change on the system.

